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A covariance proof of Gaussian Markov–Bernstein inequalities for $p \geq 2$

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Research note

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Abstract

For every $2 \leq p < \infty$, every dimension n, and every complex polynomial P on $\mathbb{R}^n$ of degree at most d, we prove $\|(x \cdot \nabla - \Delta)P\|_{L^p(\gamma_n)} \leq 4e\, d\, \|P\|_{L^p(\gamma_n)}$. The proof uses a subharmonicity property of Gaussian moments as a function of complex pseudocovariance, followed by homogenization and a complex rotation. A smooth regularization treats zeros without fractional-power branch arguments. The range $1 < p < 2$ is not proved here.

1 Statement and proof mechanism

Let γ_n be standard Gaussian measure on $\mathbb{R}^n$ and let $\mathcal{N} = x \cdot \nabla - \Delta$ be the nonnegative Ornstein–Uhlenbeck number operator. Degree always means total algebraic degree.

Theorem 1. *For $2 \leq p < \infty$, $n \geq 1$, and every polynomial $P : \mathbb{R}^n \to \mathbb{C}$ of degree at most d,*

$$\|\mathcal{N}P\|_{L^p(\gamma_n)} \leq 4e\, d\, \|P\|_{L^p(\gamma_n)}. \tag{1}$$

If P is even or odd under $x \mapsto -x$, $4e$ can be replaced by $2e$. For real-valued P, writing $g \sim \gamma_1$, we also have

$$\|\nabla P\|_{L^p(\gamma_n)} \leq \frac{2\sqrt{2e}}{\|g\|_{L^p(\gamma_1)}} \sqrt{d}\, \|P\|_{L^p(\gamma_n)}. \tag{2}$$

The factor 2 in the numerator of (2) is unnecessary when P is even or odd.

The central estimate compares a homogeneous polynomial evaluated at a complex Gaussian with its value at a real Gaussian of the same Hermitian covariance. Homogeneity makes the moment a radial function of the complex pseudocovariance. A fourth-derivative positivity identity makes that radial function subharmonic, so its largest value lies at the real Gaussian boundary. Auxiliary Gaussian squared radii then remove homogeneity. Cauchy's estimate for the analytic rotation $P(X \cos z + Y \sin z)$ produces the factor d.

Sharp degree dependence at even exponents was obtained in [2]. The all-exponent dimension-free Gaussian question is formulated in [1, Question 3 and equation (13)]. The argument below establishes the stated range $p \geq 2$.

2 A regularized fourth-derivative inequality

On $\mathbb{C}^n$, use Wirtinger derivatives and write

$$\mathscr{D} = \sum_{j=1}^{n} \partial_{z_j}^2, \qquad \overline{\mathscr{D}} = \sum_{j=1}^{n} \partial_{\bar{z}_j}^2.$$

These are pure complex second derivatives, not the usual real Laplacian.

Lemma 2. *If H is a holomorphic polynomial, $s \geq 1$, and $\varepsilon > 0$, then*

$$\mathscr{D}\overline{\mathscr{D}}(|H|^2 + \varepsilon)^s \geq 0$$

pointwise on $\mathbb{C}^n$.

Proof. Set $q = |H|^2$, $A = \sum_j (\partial_{z_j} H)^2$, $B = \mathscr{D}H$, and $f(q) = (q+\varepsilon)^s$. Direct differentiation gives

$$\mathscr{D}\overline{\mathscr{D}} f(q) = a|A|^2 + 2\,\mathrm{Re}(b\bar{H}A\bar{B}) + c|B|^2, \tag{3}$$

where

$$a = q^2 f^{(4)} + 4qf''' + 2f'', \qquad b = qf''' + 2f'', \qquad c = qf'' + f'.$$

More explicitly,

$$\begin{aligned} a &= s(s-1)(q+\varepsilon)^{s-4}\big[s(s-1)q^2 + 4(s-1)q\varepsilon + 2\varepsilon^2\big], \\ b &= s(s-1)(q+\varepsilon)^{s-3}(sq + 2\varepsilon), \\ c &= s(q+\varepsilon)^{s-2}(sq+\varepsilon). \end{aligned}$$

Thus $a, c \geq 0$, and

$$\begin{aligned} ac - qb^2 = s^2(s-1)(q+\varepsilon)^{2s-6}\varepsilon \\ \cdot \big[s(s-1)q^2 + 2sq\varepsilon + 2\varepsilon^2\big] \geq 0. \end{aligned}$$

The Hermitian quadratic form in (3) is therefore positive semidefinite. This also covers $s = 1$, when $a = b = 0$ and $c = 1$. □

3 Comparison of real and complex Gaussian moments

For $\tau = u + iv$ with $|\tau| < 1$, let $Z_\tau \in \mathbb{C}^n$ be centered Gaussian with independent coordinate pairs and

$$\mathrm{Cov}(\mathrm{Re}\, Z_{\tau,j}, \mathrm{Im}\, Z_{\tau,j}) = \frac{1}{2}\begin{pmatrix} 1+u & v \\ v & 1-u \end{pmatrix}.$$

Equivalently,

$$\mathbb{E} Z_{\tau,j}\overline{Z_{\tau,k}} = \delta_{jk}, \qquad \mathbb{E} Z_{\tau,j} Z_{\tau,k} = \tau\delta_{jk}.$$

The boundary value Z_1 is a standard real Gaussian vector.

Lemma 3. *Let $H : \mathbb{C}^n \to \mathbb{C}$ be a homogeneous holomorphic polynomial of degree D. For $p \geq 2$ and $|\tau| \leq 1$,*

$$\mathbb{E}|H(Z_\tau)|^p \leq \mathbb{E}|H(X)|^p, \qquad X \sim \gamma_n. \tag{4}$$

Proof. Fix $\varepsilon > 0$, put $s = p/2$, and define $\Phi_\varepsilon(\tau) = \mathbb{E}(|H(Z_\tau)|^2 + \varepsilon)^s$. For any smooth function ϕ with derivatives of polynomial growth, Gaussian covariance differentiation gives

$$\partial_\tau \mathbb{E}\phi(Z_\tau) = \frac{1}{2}\mathbb{E}\mathscr{D}\phi(Z_\tau), \qquad \partial_{\bar\tau}\mathbb{E}\phi(Z_\tau) = \frac{1}{2}\mathbb{E}\overline{\mathscr{D}}\phi(Z_\tau).$$

For completeness, writing $z = x + iy$, differentiation in u and v gives, respectively,

$$\frac{1}{4}\mathbb{E}(\Delta_x - \Delta_y)\phi, \qquad \frac{1}{2}\mathbb{E}\sum_j \partial_{x_j}\partial_{y_j}\phi.$$

Combining these with $\partial_\tau = (\partial_u - i\partial_v)/2$ gives the displayed identities. Lemma 2 now implies

$$\partial_\tau \partial_{\bar\tau} \Phi_\varepsilon = \frac{1}{4} \mathbb{E} \mathscr{D} \overline{\mathscr{D}} (|H|^2 + \varepsilon)^s (Z_\tau) \geq 0.$$

Hence Φ_ε is subharmonic in the unit disk. Multiplication of Z_τ by $e^{i\theta}$ changes its pseudocovariance to $e^{2i\theta}\tau$. Homogeneity gives $|H(e^{i\theta}z)| = |H(z)|$, so Φ_ε is radial. A smooth radial subharmonic function is nondecreasing in radius: if $\Phi_\varepsilon(\tau) = h(|\tau|)$, then $(rh'(r))' \geq 0$ and $rh'(r) \to 0$ at the origin.

Gaussian polynomial moments give continuity as $|\tau| \uparrow 1$. Consequently $\Phi_\varepsilon(\tau) \leq \Phi_\varepsilon(1)$. Letting $\varepsilon \downarrow 0$ proves (4). □

4 Homogenization and the complex rotation

Let X, Y be independent standard real Gaussian vectors and set

$$W_v = X \cosh v + iY \sinh v, \qquad c = \cosh(2v).$$

Then $\mathbb{E} W_{v,j} W_{v,k} = \delta_{jk}$ and $\mathbb{E} W_{v,j} \overline{W_{v,k}} = c\delta_{jk}$. Thus $W_v/\sqrt{c}$ has the law of $Z_{1/c}$. For homogeneous H of degree D, Lemma 3 gives

$$\|H(W_v)\|_p \leq c^{D/2} \|H(X)\|_p . \tag{5}$$

Now let $P = \sum_k P_k$ have a single parity, where P_k is homogeneous of degree k, $k \equiv D \pmod 2$, and $\deg P \leq D$. For $M \geq 1$, define the homogeneous polynomial of degree D

$$H_M(x, y) = \sum_k P_k(x) \left(\frac{1}{M} \sum_{j=1}^{M} y_j^2 \right)^{(D-k)/2} .$$

Apply (5) in dimension $n + M$. At real Gaussian inputs the auxiliary average $M^{-1} \sum_j y_j^2$ converges to 1 in every finite moment. The same is true at the complex Gaussian inputs W_v, because the coordinates are independent, have all moments, and satisfy $\mathbb{E} W_{v,j}^2 = 1$. Therefore both sides converge in L^p, giving

$$\|P(W_v)\|_p \leq (\cosh 2v)^{D/2} \|P(X)\|_p \leq e^{Dv^2} \|P\|_p . \tag{6}$$

The convergence assertion follows, for example, by expanding the fixed integer powers of the auxiliary average and using its uniform moment bounds together with independence from x.

For arbitrary P, write $P_e(x) = (P(x) + P(-x))/2$ and $P_o(x) = (P(x) - P(-x))/2$. Both projections are $L^p(\gamma_n)$ contractions and preserve the degree bound. Hence

$$\|P(W_v)\|_p \leq 2e^{dv^2} \|P\|_p . \tag{7}$$

Proof of Theorem 1. The case $d = 0$ is immediate. For $d \geq 1$, consider the entire, $L^p(\gamma_{2n})$-valued function

$$F(z) = P(X \cos z + Y \sin z).$$

Its values lie in a finite-dimensional space of polynomials in (X, Y), so Banach-valued Cauchy estimates apply directly. For $z = u + iv$, real Gaussian rotational invariance and (7) give $\|F(z)\|_p \leq 2e^{d(\operatorname{Im} z)^2} \|P\|_p$. On $|z| = d^{-1/2}$ this yields

$$\|F''(0)\|_p \leq 4e\, d \|P\|_p .$$

But

$$F''(0) = Y^{\mathsf{T}}\nabla^2 P(X)Y - X\cdot\nabla P(X), \qquad \mathbb{E}_Y F''(0) = -\mathcal{N}P(X).$$

Conditional-expectation contraction proves (1). For real P, use $F'(0) = Y\cdot\nabla P(X)$ and radius $(2d)^{-1/2}$ instead. Since $\|F'(0)\|_p = \|g\|_p \|\nabla P\|_p$, this gives (2). For a single parity, use (6) throughout and omit the factor 2. □

5 The remaining range $1 < p < 2$

The covariance comparison used above is false below 2, already for $H(z) = z$. Indeed, if X, Y are independent standard real Gaussians and $1 < p < 2$, strict concavity of $t \mapsto t^{p/2}$ gives

$$\mathbb{E}\left|\frac{X+iY}{\sqrt{2}}\right|^p = \mathbb{E}\left(\frac{X^2+Y^2}{2}\right)^{p/2} > \mathbb{E}|X|^p.$$

This disproves that auxiliary comparison in the missing range, not the desired Markov–Bernstein inequality.

A sufficient remaining estimate, for real polynomials, is

$$\int |P|^{p-2}|\nabla P|^2\, d\gamma_n \le B_p d \int |P|^p\, d\gamma_n, \qquad 1 < p < 2, \tag{8}$$

with B_p independent of n, d. To see sufficiency, assume $P \neq 0$ and $d \ge 1$ and set $P_t = e^{t\mathcal{N}}P$, which remains a polynomial of degree at most d. Gaussian integration by parts gives

$$\frac{d}{dt}\log\|P_t\|_p = (p-1)\frac{\int |P_t|^{p-2}|\nabla P_t|^2\, d\gamma_n}{\int |P_t|^p\, d\gamma_n} \le (p-1)B_p d.$$

Thus $\left\|e^{t\mathcal{N}}P\right\|_p \le e^{(p-1)B_p dt}\|P\|_p$. Combining this with the dimension-free analytic-semigroup estimate $\left\|\mathcal{N}e^{-t\mathcal{N}}\right\|_{L^p\to L^p} \le A_p/t$ and taking $t = ((p-1)B_p d)^{-1}$ yields the desired linear-degree generator bound. Conversely, that generator bound implies (8) by integration by parts and Hölder's inequality. Therefore (8) isolates the unresolved part, rather than replacing it with an already proved fact.

References

[1] A. Eskenazis and P. Ivanisvili, *Dimension independent Bernstein–Markov inequalities in Gauss space*, arXiv:1808.01273.

[2] E. Kosov, *Dimension-free Markov–Bernstein inequalities for product measures*, arXiv:2606.13575, 2026.